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[Intervention Review]

Atherectomy for peripheral arterial disease

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ABSTRACT

Rationale

Peripheral arterial disease (PAD) is a condition most commonly caused by atherosclerotic narrowing of lower limb arteries, resulting in intermittent claudication, chronic limb-threatening ischaemia or acute limb ischaemia. There are various treatment strategies, including atherectomy, a technique used during endovascular surgery where the atheroma is cut or ground away within the artery. Another procedure, such as balloon angioplasty, is often performed at the same time. The studies investigating atherectomy for PAD have all been small-scale, with varying methodologies and, as a result, it is unclear if atherectomy is a more effective treatment for PAD compared to more conventional treatments. Despite this, rates of atherectomy use are increasing, especially in the United States. This review focuses on randomised controlled trials and is the second update of a Cochrane review, following the original publication in 2014 and the first update in 2020.

Objectives

To evaluate the benefits and harms of atherectomy as a treatment for peripheral arterial disease compared to other treatments.

Search methods

We searched the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, Embase, Cumulative Index to Nursing and Allied Health Literature (CINAHL) and Allied and Complementary Medicine (AMED) databases, and the World Health Organization International Clinical Trials Registry Platform and ClinicalTrials.gov trials registers from 13 August 2019 to 28 January 2025.

Eligibility criteria

We included all randomised controlled trials that compared atherectomy with other established treatments. All participants had symptomatic PAD with either claudication or chronic limb-threatening ischaemia and evidence of atherosclerotic lower limb arterial disease.

Outcomes

Outcomes of interest were: primary patency, all-cause mortality, fatal and non-fatal cardiovascular events, target vessel revascularisation rates and complication rates.

Risk of bias

We used the Cochrane risk of bias tool (RoB 1) to assess the risk of bias in the studies. We judged all included studies to have a high risk of overall bias.

Synthesis methods

Two review authors screened studies for inclusion, extracted data, assessed risk of bias and used the GRADE criteria to assess the certainty of the evidence. Any disagreements were resolved through discussion. We synthesised results for each outcome using meta-analysis where possible (random-effects model, dichotomous outcomes assessed using the Mantel-Haenszel method, continuous outcomes assessed using the inverse variance method).

Included studies

We included 11 studies, with a total of 814 participants and 872 treated lesions.

Synthesis of results

We found two comparisons: atherectomy versus balloon angioplasty (atherectomy versus BA) and atherectomy versus BA with primary stenting (atherectomy versus stenting). No studies compared atherectomy with bypass surgery. Overall, the evidence from this review was of very low certainty, due to a high risk of bias, imprecision and inconsistency.

Ten studies (659 participants, 717 treated lesions) compared atherectomy versus BA. There was no evidence of differences between atherectomy and BA for the primary outcomes: six-month primary patency rates (risk ratio (RR) 1.24, 95% confidence interval (CI) 0.92 to 1.68; 6 studies, 298 participants; very low-certainty evidence); 12-month primary patency rates (RR 1.13, 95% CI 0.96 to 1.34; 5 studies, 326 participants; very low-certainty evidence); mortality rates (RR 0.50, 95% CI 0.24 to 1.02; 7 studies, 493 participants; very low-certainty evidence) or cardiovascular events at 12 months (RR 0.59, 95% CI 0.13 to 2.70; 2 studies, 163 participants; very low-certainty evidence).

There was no evidence of differences when examining: six-month target vessel revascularisation (TVR) rates (RR 0.61, 95% CI 0.24 to 1.56; 5 studies, 348 treated vessels; very low-certainty evidence), 12-month TVR (RR 0.68, 95% CI 0.41 to 1.12; 6 studies, 371 treated vessels; very low-certainty evidence) or complication rates (RR 0.84, 95% CI 0.34 to 2.04; 7 studies, 457 participants; very low-certainty evidence).

One study (155 participants, 155 treated lesions) compared atherectomy versus stenting, so the comparison was extremely limited and subject to imprecision. This study did not report primary patency. There was no evidence of a difference in the atherectomy versus stenting arms for mortality rates (RR 0.38, 95% CI 0.04 to 3.23; 155 participants; very low-certainty evidence), cardiovascular events (RR 0.38, 95% CI 0.04 to 3.23; 155 participants; very low-certainty evidence) and TVR rates at six months (RR 2.27, 95% CI 0.95 to 5.46; 155 participants; very low-certainty evidence). The study did not report on TVR at 12 months. There was no evidence of a difference in complication rates between the two arms (RR 7.04, 95% CI 0.80 to 62.23; 155 participants; very low-certainty evidence).

There are several limitations to the evidence. The studies were of small sample size, with poor methodological quality, considerable variations in protocols and a high overall risk of bias due to high attrition and a lack of blinding.

Authors' conclusions

This review update shows that the evidence is still very uncertain about the effect of atherectomy on primary patency, mortality and cardiovascular event rates compared to plain balloon angioplasty with or without stenting alone. We identified no evidence of differences in target vessel revascularisation rates and complication rates, although this is again uncertain. The included studies were small, heterogeneous and at high risk of bias. Larger studies that are powered to detect clinically meaningful, patient-centred outcomes are required.

Funding

This Cochrane review had no dedicated funding.

Registration

Protocol and previous versions available via 10.1002/14651858.CD006680, 10.1002/14651858.CD006680.pub3.

PLAIN LANGUAGE SUMMARY

Is atherectomy (cutting or grinding away the hardened fatty plaques) in leg arteries better than other minimally invasive procedures for peripheral arterial disease?

Key messages

- There seems to be little to no difference in either benefits or harms when the use of atherectomy (a minimally invasive procedure that uses a device to cut or grind away the hardened fatty plaques) is compared to conventional treatments, including balloon angioplasty

(a balloon is used to stretch the narrowed or blocked artery) with or without stenting (a tube used to hold open the segment of artery). However, this is based on poor-quality evidence and hence the findings from this review are very uncertain.

- Larger, better-designed studies are needed to better evaluate the use of atherectomy compared to other conventional treatments.

What is peripheral arterial disease?

Peripheral arterial disease is a common condition in which fatty plaques cause narrowing or blockage of the arteries of the legs. Individuals with this condition may experience pain while walking, pain at rest or leg ulceration due to inadequate blood supply. The number of people with peripheral arterial disease is increasing globally.

How is this condition treated?

Treatment options are walking therapy, open surgery using a blood vessel or graft to bypass the narrowed or blocked section of the artery, or minimally invasive surgery. The options for minimally invasive surgery include balloon angioplasty, when a deflated balloon is passed into the narrowing over a wire, then blown up to stretch the artery, and stenting (used in addition to balloon angioplasty), which holds open the balloon-stretched section for extra support.

A final option, less commonly used, is a technique called atherectomy. This treatment uses a device that cuts or grinds away the hardened deposition (atheroma) within the artery. Atherectomy is less well researched compared to the other treatment methods. Despite this, the use of atherectomy is increasing.

What did we want to find out?

The main question we wanted to evaluate was whether atherectomy was better than balloon angioplasty, stenting or open bypass surgery for treating people with peripheral arterial disease. We also wanted to find out if atherectomy was better than balloon angioplasty for peripheral arterial disease located in the leg, above the knee and whether the use of drug-releasing balloons or stents had any effects on outcomes.

What did we do?

We searched for studies that compared atherectomy to any of the other methods for treating peripheral arterial disease. We combined the results of these studies with the studies already identified in the two previous versions of this review. We rated our overall confidence in the studies based on their design and size.

What did we find?

We found four new studies that compared atherectomy to other minimally invasive treatment options (balloon angioplasty with or without stenting), to make a total of 11 studies, with a total of 814 participants, included in this review. Only one study compared atherectomy to stenting, with the remainder comparing atherectomy to balloon angioplasty. The majority of studies were either performed in Europe or the United States, with the remaining study from China. We did not find any studies comparing atherectomy to open bypass surgery.

From evaluating the data from the studies identified, there seems to be little to no difference between atherectomy and balloon angioplasty or primary stenting when looking at the following outcomes: maintaining vessel patency (openness) at 6 and 12 months, the risk of death, the risk of subsequent disease or injury to the heart or blood vessels, the likelihood of needing repeat treatment on the target artery or the overall risk of complications.

What are the limitations of the evidence?

We are not confident in the evidence because, in several of the studies, it is possible that the participants and assessors were aware of which treatment individuals were receiving. Furthermore, not all participants were followed up for the entire length of the studies, and not all of the studies provided data about everything we were interested in.

How up-to-date is this evidence?

This review is an update of previous Cochrane reviews published in 2014 and 2020. The evidence is up-to-date as of January 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Atherectomy compared to balloon angioplasty for peripheral arterial disease

Atherectomy compared to balloon angioplasty for peripheral arterial disease

Patient or population: people with PAD
Setting: hospital setting in Europe, North America and China
Intervention: atherectomy
Comparison: BA

Outcomes	Nº of participants (studies)	Certainty of the evidence (GRADE)	Relative effect (95% CI)	Anticipated absolute effects* (95% CI)	
				Risk with BA	Risk with atherectomy
Primary patency (follow-up: 6 months)	298 (6 RCTs)	⊕⊕⊕⊕ VERY LOW ^a	RR 1.24 (0.92 to 1.68)	504 per 1000	625 per 1000 (463 to 846)
Primary patency (follow-up: 12 months)	326 (5 RCTs)	⊕⊕⊕⊕ VERY LOW ^a	RR 1.13 (0.96 to 1.34)	633 per 1000	715 per 1000 (608 to 848)
Mortality (follow-up: 12 months)	493 (7 RCTs)	⊕⊕⊕⊕ VERY LOW ^b	RR 0.50 (0.24 to 1.02)	101 per 1000	51 per 1000 (24 to 103)
Fatal and non-fatal cardiovascular events (follow-up: 24 months)	163 (2 RCTs)	⊕⊕⊕⊕ VERY LOW ^b	RR 0.59 (0.13 to 2.70)	48 per 1000	28 per 1000 (6 to 129)
TVR (follow-up: 6 months)	348 (5 RCTs)	⊕⊕⊕⊕ VERY LOW ^b	RR 0.61 (0.24 to 1.56)	64 per 1000	39 per 1000 (15 to 100)
TVR (follow-up: 12 months)	371 (6 RCTs)	⊕⊕⊕⊕ VERY LOW ^b	RR 0.68 (0.41 to 1.12)	151 per 1000	103 per 1000 (62 to 170)
Complication rates (follow-up: 12 months)	457 (7 RCTs)	⊕⊕⊕⊕ VERY LOW ^{b,c}	RR 0.84 (0.34 to 2.04)	186 per 1000	157 per 1000 (63 to 380)

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

BA: balloon angioplasty; **CI:** confidence interval; **PAD:** peripheral arterial disease; **RCT:** randomised controlled trial; **RR:** risk ratio; **TVR:** target vessel revascularisation

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect

^aWe downgraded primary patency at 6 and 12 months by three levels: two levels due to risk of bias, as high risk of bias was identified across a number of criteria for all studies included (lack of blinding and high rates of attrition), and one level due to imprecision (95% CI including a risk ratio of 1.0 and including appreciable benefit).

^bWe downgraded mortality, fatal and non-fatal cardiovascular events, TVR at 6 and 12 months, and complication rates by four levels: two levels due to risk of bias, as high risk of bias was identified across a number of criteria for all studies included (lack of blinding and high rates of attrition) and two levels due to imprecision (small trials, with few events, and wide confidence intervals including risk ratio of 1.0 and including appreciable benefit and/or harm).

^cWe also downgraded complication rates by one level due to inconsistency as the P value met the criteria for significant heterogeneity.

Summary of findings 2. Atherectomy compared to balloon angioplasty with primary stenting for peripheral arterial disease

Atherectomy compared to balloon angioplasty with primary stenting for peripheral arterial disease

Patient or population: people with PAD

Setting: hospital setting in Europe

Intervention: atherectomy

Comparison: BA with primary stenting

Outcomes	Nº of participants (studies)	Certainty of the evidence (GRADE)	Relative effect (95% CI)	Anticipated absolute effects* (95% CI)		Comments
Primary patency (follow-up 6 months)	Not reported for this comparison					—
Primary patency (follow-up 12 months)	Not reported for this comparison					—
Mortality (follow-up: 24 months)	155 (1 RCT)	⊕⊕⊕⊕ VERY LOW ^a	RR 0.38 (0.04 to 3.23)	40 per 1000	15 per 1000 (2 to 129)	—
Fatal and non-fatal cardiovascular events	155 (1 RCT)	⊕⊕⊕⊕ VERY LOW ^a	RR 0.38 (0.04 to 3.23)	40 per 1000	15 per 1000 (2 to 129)	Ott 2017 reported 4 deaths at 24 months (3 deaths in the drug-eluting balloon and

(follow-up: 24 months)						stent arm and 1 death in the plain balloon and stent arm), which they attributed to underlying cardiovascular disease, but no specific causes were stated.
TVR (follow-up: 6 months)	155 (1 RCT)	⊕⊕⊕⊕ VERY LOW ^a	RR 2.27 (0.95 to 5.46)	80 per 1000	182 per 1000 (76 to 437)	—
TVR (follow-up: 12 months)	Not reported for this comparison.					—
Complication rates (follow-up: 24 months)	155 (1 RCT)	⊕⊕⊕⊕ VERY LOW ^a	RR 7.04 (0.80 to 62.23)	See footnote ^b		Ott 2017 reported 3 complications, all 3 of which were in the atherectomy arm: 2 vessel perforations and 1 flow-limiting dissection.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

BA: balloon angioplasty; **CI:** confidence interval; **PAD:** peripheral arterial disease; **RCT:** randomised controlled trial; **RR:** risk ratio; **TVR:** target vessel revascularisation

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect

Moderate certainty: we are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different

Low certainty: our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect

Very low certainty: we have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

^aWe downgraded by three levels: one level due to risk of bias (inadequate blinding and high rates of attrition) and two levels due to imprecision (small trial with few events and wide confidence intervals including a risk ratio of 1.0 and including appreciable benefit and/or harm).

^bAbsolute effects could not be meaningfully calculated for the complication rates outcome because there were no events in the control group. Observed risks: 3 per 55 participants (intervention) versus 0 per 100 participants (comparison).

BACKGROUND

Description of the condition

Peripheral arterial disease (PAD) refers to the lower limb manifestation of systemic atherosclerotic disease. Atherosclerotic plaque deposits result in narrowing or complete occlusion of the lumen of the arteries supplying the lower limb. Clinical manifestations can range from asymptomatic to pain on exertion (intermittent claudication) to rest pain with tissue loss or gangrene (chronic limb-threatening ischaemia) [1]. The global burden of disease 2021 analysis identified an almost doubling of PAD cases with an estimated global prevalence of 113.7 million people and a peak incidence of 75 to 79 years of age [2]. The condition is expected to rise by 220% to an estimated prevalence of 360 million by 2050 [3]. There has also been a shift in the most prevalent risk factors, with high fasting plasma glucose now being the highest contributor, whilst the impact of smoking has declined [2]. There are several options for the treatment of symptomatic peripheral arterial disease (PAD), including exercise therapy, endovascular surgery or bypass surgery [4, 5].

Description of the intervention and how it might work

The purpose of atherectomy is the removal of plaque and calcified tissue. The plaque is debulked, thereby increasing the diameter of the vessel lumen. In contrast, conventional endovascular treatments of angioplasty and stenting involve disruption and displacement of plaque against the vessel wall. Atherectomy may be advantageous in highly calcified lesions, where angioplasty may lead to vessel recoil and may be less likely to result in dissection as a result of barotrauma.

In atherectomy, the method of plaque debulking is variable, giving rise to four main types of atherectomy: laser, directional, rotational and orbital [6, 7]. Laser atherectomy uses a laser tip catheter to ablate the atheroma. Directional atherectomy involves the use of a rotating cutting blade, usually on one side of the catheter, which can be directed to the lesion and rotated in the preferred direction to cut and remove atheromatous tissue. Tissue is collected into a nosecone, which needs to be emptied. There is generally no active aspiration mechanism and, as a result, embolic protection filter use is advisable. Rotational atherectomy is similar to directional, but with the rotating burr at the tip of the catheter device. The devices available for rotational atherectomy generally have a continuous aspiration mechanism. Orbital atherectomy involves debulking via a diamond-coated crown at the end of the catheter, which orbits, producing circumferential plaque resection. Generally, the devices available do not have an aspiration function, so an embolic protection filter is used.

Why it is important to do this review

This is the second update of a Cochrane review first published in 2014, which included four trials with small numbers of participants [8]. The first update published in 2020 included seven trials [9]. The number of studies that met the inclusion criteria was low, and they were of poor methodological quality.

Despite the lack of robust evidence, atherectomy use has soared, particularly in the United States. This has been well documented in the literature, with the publication of retrospective analyses of registry data. A 2021 retrospective review of Medicare fee-for-service claims identified 58,552 patients who underwent

intervention for PAD, with 53.8% of this cohort undergoing atherectomy as the index procedure [10]. Particularly concerning from this analysis were the characteristics associated with atherectomy use. These included patients with claudication, physicians working in office-based laboratory settings and non-vascular surgery specialities. A secondary analysis of the registry data identified a significant increase in re-intervention rates for those patients treated with atherectomy for claudication [11]. Another study reviewing registry data from 2010 to 2015, including 16,338 patients, compared outcomes of atherectomy use to angioplasty with or without stenting [12]. The authors identified five-year rates of amputation and major adverse limb events as significantly higher in the atherectomy cohort.

The concerning practice patterns surrounding the overuse of atherectomy has resulted in media attention and significant controversy, with the publication of a *New York Times* article providing an in-depth investigation into atherectomy practices [13]. The article identified a network of outpatient clinics, incentivised by medical device companies to perform atherectomy procedures, even in patients with asymptomatic PAD or claudication. The Society of Vascular Surgery responded to the article, expressing concern over the misuse of invasive treatment, discouraging intervention in claudication and stating that most vascular surgeons do follow evidence-based practices [14].

Since the publication of the previous Cochrane review in 2020, further systematic reviews and meta-analyses have been published in the literature. However, these have had conflicting results and include observational studies within the meta-analyses [15, 16]. In light of the concerns around atherectomy use and the clinical uncertainty regarding its effectiveness, there is a need for an up-to-date systematic review and meta-analysis of the best available evidence investigating the use of atherectomy for patients with PAD.

OBJECTIVES

To evaluate the benefits and harms of atherectomy as a treatment for peripheral arterial disease compared to other treatments.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review and PRISMA 2020 for the reporting. We assessed the quality of the trials using Cochrane's risk of bias tool (RoB 1) [17]. This tool has since been updated; however, the previous one was used to maintain consistency with the previous version of the review [18]. We used the GRADE approach to assess the certainty of the evidence.

Differences from the protocol and previous versions of the review

The Cochrane Vascular Specialised Register via the Cochrane Register of Studies (CRS-Web) was not searched for this review. As part of this update, we have increased the number of summary of findings tables from two to four. The two new tables include the results of the subgroup analysis investigating the use of drug-eluting technology. We have included these as all four new included studies used drug-eluting technology when performing balloon angioplasty.

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) comparing atherectomy with other established treatments, including angioplasty, stenting and bypass surgery.

Types of participants

We included participants with symptomatic peripheral arterial disease (PAD) with either claudication or chronic limb-threatening ischaemia and evidence of lower limb arterial disease. We considered arterial disease in any peripheral territory. We excluded studies with participants who had previously had bypass surgery, percutaneous transluminal angioplasty (PTA) or stents in the target lesion, as these treatments might affect the primary patency rates. If studies had included a subset of eligible patients, data from this cohort would have been added to our meta-analysis. If we were unable to isolate data relating to the eligible participants, this study would have been excluded.

Types of interventions

We included RCTs that compared atherectomy against any established treatment for PAD, in order to evaluate the effectiveness of atherectomy. Eligible co-interventions included atherectomy in combination with balloon angioplasty or stenting. We identified the following comparisons from the inclusion criteria.

Atherectomy plus balloon angioplasty versus balloon angioplasty (referred to as atherectomy versus BA)

Subgroups:

- Atherectomy plus plain balloon angioplasty versus plain balloon angioplasty (referred to as atherectomy versus PBA)
- Atherectomy plus drug-eluting balloon angioplasty versus drug-eluting balloon angioplasty (referred to as atherectomy plus DEB versus DEB)

Atherectomy versus primary stenting (referred to as atherectomy versus stenting)

Subgroups

- Atherectomy versus drug-eluting balloon angioplasty plus primary stenting (referred to as atherectomy versus DEB plus stenting)
- Atherectomy versus plain balloon angioplasty plus primary stenting (referred to as atherectomy versus PBA plus stenting).

Atherectomy versus surgical bypass procedure

Outcome measures

We did not use the outcomes to formulate exclusion/inclusion criteria, but they were instead outcomes of interest to be identified from the included studies.

Critical outcomes

- Primary vessel patency, as assessed by ankle brachial index (ABI), arterial doppler ultrasound or angiography at six months and one year, and as data available in the studies.

- All-cause mortality at six months and one year, and as data available in the studies.
- Fatal and non-fatal cardiovascular events at six months and one year, and as data available in the studies.

Important outcomes

- Immediate procedural and angiographic outcomes including:
 - technical failure rates (the rate of failure of the procedure to perform the desired outcome of patent vessel with blood flow);
 - bailout stenting;
 - balloon inflation pressures.
- Target vessel revascularisation rates (TVR) (defined as any repeat intervention required to the target vessel).
- Complication rates, including thrombus, embolus, perforation and aneurysm.
- Morbidity assessment, including:
 - tissue healing;
 - avoidance of any amputation; and
 - performance of less extensive amputation.
- Quality of life (QoL) outcomes, as measured in the included studies.
- Clinical and symptomatic outcomes, e.g. improved walking distance, symptom relief.

Search methods for identification of studies

Electronic searches

The authors conducted systematic searches of the following databases for randomised controlled trials and controlled clinical trials without language, publication year or publication status restrictions ([Supplementary material 1](#)):

- Cochrane Central Register of Controlled Trials (CENTRAL) via the Cochrane Register of Studies Online (CRSO);
- MEDLINE (Ovid MEDLINE Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE Daily and Ovid MEDLINE) 1946 to present (searched from 13 August 2019 to 28 January 2025);
- Embase Ovid (searched from 13 August 2019 to 28 January 2025);
- CINAHL EBSCO (Cumulative Index to Nursing and Allied Health Literature; searched from 13 August 2019 to 28 January 2025);
- AMED Ovid (Allied and Complementary Medicine Database; searched from 13 August 2019 to 28 January 2025).

The authors searched the following trial registries on 28 January 2025:

- World Health Organization International Clinical Trials Registry Platform (who.int/trialsearch);
- ClinicalTrials.gov (clinicaltrials.gov).

We used the Cochrane specialised filters for searching for randomised controlled trials in the databases Embase and MEDLINE [18].

Searching other resources

We also searched the reference lists of the included studies and relevant systematic reviews to identify studies to include. We searched for errata or retractions for included studies published on PubMed, Retraction Watch and journal websites.

Data collection and analysis

Selection of studies

Two review authors (SP and AG) independently selected trials for inclusion in the review. They resolved any disagreements through discussion. The section [Criteria for considering studies for this review](#) details the inclusion criteria used for the selection process. The results of the search were managed through Mendeley software and Excel.

Data extraction and management

SP extracted the data, and CT cross-checked them. They resolved any disagreements through discussion. The data extraction form was not piloted. SP extracted the following information for each trial.

- Trial methods: method of randomisation, method of allocation.
- Participants: country of origin, age, sex distribution, severity of disease, as measured by the ABI and using the European Consensus definition of critical ischaemia, inclusion and exclusion criteria [19].
- Interventions: type of procedure (atherectomy, angioplasty or bypass).
- Outcomes: primary and secondary outcomes, as listed in [Outcome measures](#).

We extracted data directly from the published papers and did not make any attempt to obtain additional unpublished data. We based all analyses on endpoint data from the individual clinical trials, which all provided intention-to-treat results. We synthesised the data by comparing group results and did not amalgamate individual participant data from different trials.

Risk of bias assessment in included studies

Two review authors (SP, AG) assessed the included studies' risk of bias independently, using the Cochrane risk of bias tool (RoB 1), according to the guidelines given in the *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1 [17]. Any disagreements were resolved through discussion.

The review authors assessed the following domains as 'low risk of bias', 'unclear risk of bias' or 'high risk of bias':

- sequence generation;
- allocation concealment;
- blinding of personnel and participants;
- blinding of outcome assessors;
- incomplete outcome data;
- selective outcome reporting; and
- other risk of bias.

The 'Characteristics of included studies' table reports the assessments for each individual study ([Supplementary material 2](#)).

Measures of treatment effect

We measured the treatment effects for dichotomous outcomes using the risk ratio (RR) with 95% confidence interval (CI). For continuous outcomes, we measured treatment effects as the mean difference (MD) with 95% CI.

Unit of analysis issues

For the outcomes of mortality, fatal and non-fatal cardiovascular events, complications, quality of life, and clinical and symptomatic outcomes, the unit of analysis was the individual participant rather than the treated vessel. Four trials included multiple treated vessels per participant in some cases (Dattilo 2014 [20]; Shammass 2011 [21, 22, 23, 24]; Shammass 2012 [25, 26, 27]; Zeller 2022 [28]). This means that the observations from these trials will not be totally independent, and therefore should have less emphasis placed on them in the meta-analysis. However, as the majority of participants in these trials had only one treated vessel, and very few had more than one treated vessel, it is not likely that this will have a large impact on the results presented below. We therefore did not feel it was necessary to introduce more sophisticated statistical methods, such as meta-regression, to take account of these differences. We could not re-examine the data at an individual participant level.

One of the trials included was multi-armed (Ott 2017 [29, 30]). The atherectomy group was split equally between the DEB plus stenting and PBA plus stenting comparisons to prevent double counting of participants.

We had planned to handle the analysis of cluster trials as per the *Cochrane Handbook* guidelines in section 21.1.3, although none were identified in this review [18].

Dealing with missing data

We performed analysis on a complete case basis, and it was not necessary to contact authors for additional data.

Reporting bias assessment

We planned to assess the likelihood of potential publication bias using funnel plots; however, due to the overall low number of studies and high heterogeneity between them, we concluded that the use of funnel plots would not be appropriate [17].

Synthesis methods

We intended to pool data from all studies when the clinical procedures followed were comparable. Where possible, we used inverse-variance random-effects models for data synthesis because the included studies used different devices for atherectomy (clinical heterogeneity) [31]. We used RevMan 7.2.0 software to synthesise the data [32].

We looked for clinical heterogeneity by examination of the study details, and we used the Chi² test to assess heterogeneity between trials, using P values less than 0.1 to indicate the possible presence of significant heterogeneity. Since trials contained low participant numbers, the power of this test is likely to be low if a small P value is used [17].

Investigation of heterogeneity and subgroup analysis

We had planned to carry out subgroup analyses where the studies reported the presence or absence of concomitant illnesses such as

diabetes, hypertension, hyperlipidaemia or chronic kidney disease. We had also planned to conduct subgroup analyses if the studies reported data on smoking, gender of participants, lesion location, length and percentage of stenosis, including whether any studies classified lesion length and percentage of stenosis according to the Trans-Atlantic Inter-Society Consensus Document on Management of Peripheral Arterial Disease (TASC II) [33]. However, the included studies did not report data on these subgroups.

We performed subgroup analysis to investigate the differences between:

- atherectomy versus plain balloon angioplasty or drug-eluting angioplasty;
- atherectomy versus drug-eluting stent plus angioplasty or plain stenting plus angioplasty.

We performed this subgroup analysis given the concerns regarding paclitaxel-eluting devices and their potential correlation with increased risk of mortality, which arose from a systematic review and meta-analysis [34]. There was significant heterogeneity between the subgroups for plain balloon angioplasty/stenting versus drug-eluting balloon angioplasty/stenting in both groups, so we used random-effects models to calculate the risk ratios.

In response to reviewer feedback, we performed a further subgroup analysis to investigate the differences between atherectomy versus balloon angioplasty specifically for femoropopliteal lesions. This was as a result of a 2023 systematic review and meta-analysis of randomised controlled trials that concluded that atherectomy use resulted in improved primary patency, reduced the major amputation rate and reduced bailout stenting rates for femoropopliteal lesions [15].

Equity-related assessment

We did not investigate health inequity in this review. The focus of this review is the direct outcomes related to the benefits and harm of using atherectomy compared to other methods of treating PAD. Equity-focused analysis is outside the scope of this review.

Sensitivity analysis

Many participants in the atherectomy arm of the included studies underwent additional angioplasty. Not all studies specified details of this exactly, so we were unable to analyse these participants separately. The result of atherectomy is still considered successful

even with additional angioplasty, so we included these participants in the atherectomy arm for analysis. Only one trial did not perform routine angioplasty with atherectomy (Vroegindeweij 1995 [35, 36, 37, 38, 39]). We performed a sensitivity analysis to assess the effect of including this study in the overall meta-analyses of the primary outcomes.

Certainty of the evidence assessment

We included summary of findings tables in this update to present the most important findings and the certainty of the evidence for the most clinically relevant outcomes. The seven outcomes in the summary of findings tables are: primary patency (six and 12 months); mortality; fatal and non-fatal cardiovascular events; TVR (six and 12 months); and complications.

We included one summary of findings table for the comparison 'Atherectomy compared to balloon angioplasty for peripheral arterial disease' ([Summary of findings 1](#)) and one for 'Atherectomy compared to balloon angioplasty with primary stenting for peripheral arterial disease' ([Summary of findings 2](#)). We also included summary of findings tables for the drug-eluting subgroup comparisons of 'Atherectomy compared to drug-eluting balloon angioplasty for peripheral arterial disease' ([Supplementary material 8](#)) and 'Atherectomy compared to drug-eluting balloon angioplasty and primary stenting for peripheral arterial disease' ([Supplementary material 9](#)).

Two review authors (SP and AG) determined the certainty of the evidence for each outcome using the GRADE approach, which considers the overall risk of bias of the included studies, the directness of the evidence, inconsistency within the results, precision of the estimate and risk of publication bias [40]. Any disagreements were resolved through discussion. We created the summary of findings tables using GRADEpro GDT software.

Consumer involvement

We did not include consumer involvement in this review, due to time and resource constraints.

RESULTS

Description of studies

See [Figure 1](#), [Table 1](#), [Supplementary material 2](#), [Supplementary material 3](#) and [Supplementary material 4](#).

Figure 1. Study flow diagram.

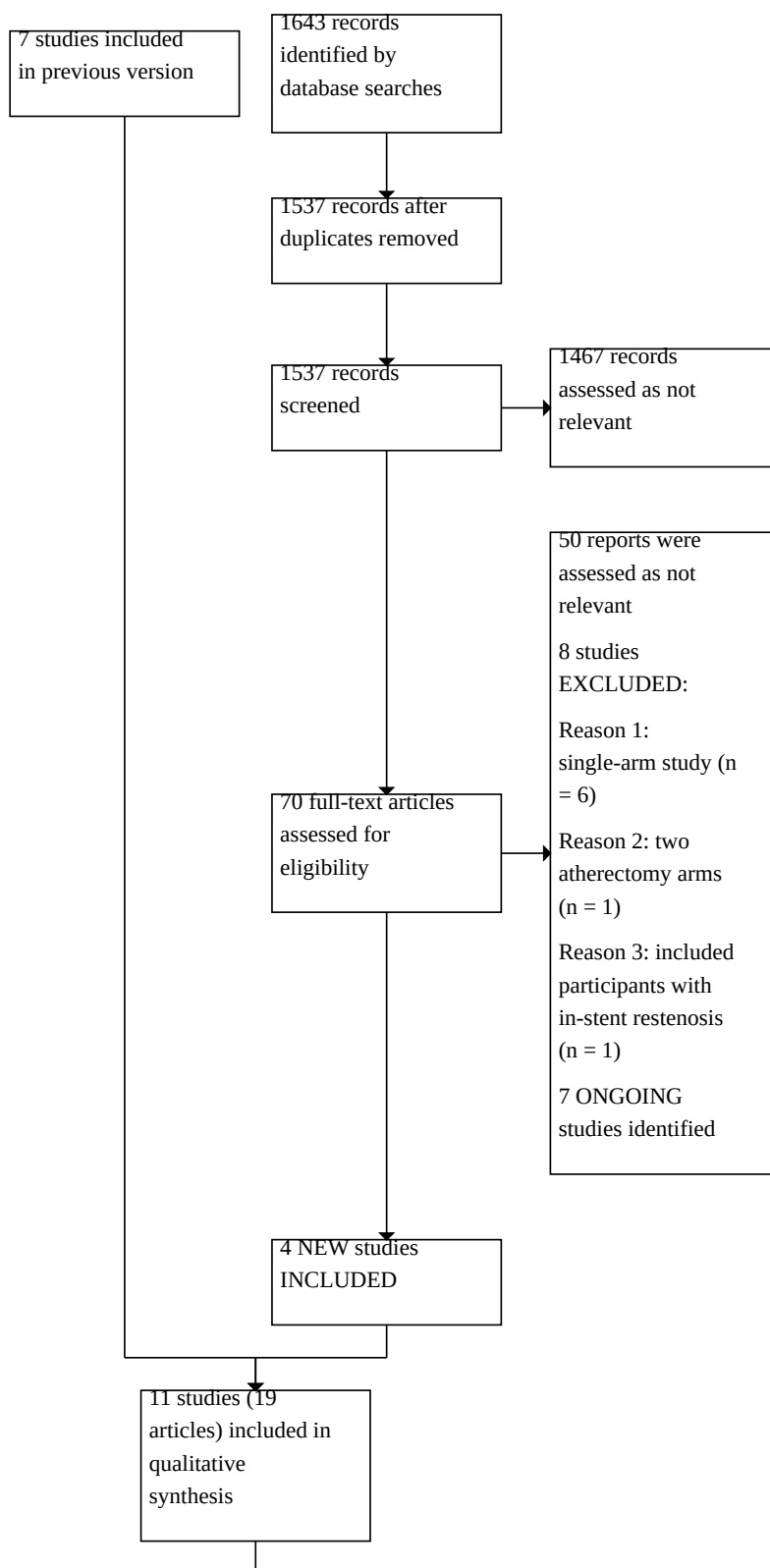
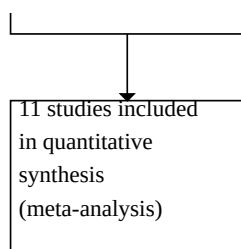


Figure 1. (Continued)



Results of the search

The search was run twice, initially on 23 December 2023 with an updated search on 28 January 2025. In total, 1643 records were identified, with a total of 1537 after duplicates were removed. We assessed 70 full-text articles for eligibility. For this update, we included four studies, excluded eight studies and identified seven studies as ongoing (Figure 1).

Included studies

The 'Characteristics of included studies' table summarises the details of the included studies (Supplementary material 2).

We identified four new studies for this update (Cai 2020 [41]; Rastan 2021 [42]; Shamma 2022 [43]; Zeller 2022). In total, 11 studies including 814 participants and 872 treated lesions met the inclusion criteria (Cai 2020; Dattilo 2014; Nakamura 1995 [44]; Ott 2017; Rastan 2021; Shamma 2011; Shamma 2012; Shamma 2022; Vroegindewij 1995; Zeller 2017 [45, 46]; Zeller 2022). All studies were prospective randomised controlled trials. The mean number of study participants was 74 (range 39 to 155 participants).

Five studies were based in the United States of America (Dattilo 2014; Nakamura 1995; Shamma 2011; Shamma 2012; Shamma 2022), one in China (Cai 2020) and the remainder in Europe (Ott 2017; Rastan 2021; Vroegindewij 1995; Zeller 2017; Zeller 2022).

Three studies had independent funding (Cai 2020; Nakamura 1995; Ott 2017), one study had an unclear source of funding (Vroegindewij 1995) and the remaining studies were funded by device manufacturers (Dattilo 2014; Rastan 2021; Shamma 2011; Shamma 2012; Shamma 2022; Zeller 2017; Zeller 2022).

Ten studies compared atherectomy to balloon angioplasty (BA) (Cai 2020; Dattilo 2014; Nakamura 1995; Rastan 2021; Shamma 2011; Shamma 2012; Shamma 2022; Vroegindewij 1995; Zeller 2017; Zeller 2022) and one study compared atherectomy to BA and primary stenting (Ott 2017). None of the studies used atherectomy followed by stenting as a primary intervention. None of the studies compared atherectomy to bypass surgery. Intervention details for each study are described in Table 2.

The outcomes investigated by each study are described in Table 1.

Excluded studies

We excluded an additional eight studies following the most recent search (Babaev 2022 [47]; Bohme 2021 [48]; Donas 2023 [49]; Kim 2023 [50]; Lida 2022 [51]; Rundback 2019 [52]; Shamma 2020 [53]; Shamma 2021 [54]), bringing the total number of excluded studies to 17 (Babaev 2022; Bohme 2021; Brodmann 2013 [55]; Del Giudice 2014 [56]; Dippel 2015 [57, 58]; Donas 2023; Gabrielli 2012 [59]; Gandini 2013 [60]; Gisbertz 2009 [61, 62]; Kim 2023; Lida 2022; NCT02730234 [63]; NCT02832024 [64]; Rundback 2019; Schwindt 2017 [65]; Shamma 2020; Shamma 2021). Two studies performed endarterectomy rather than atherectomy (Gabrielli 2012; Gisbertz 2009). Six studies were excluded as the participants involved had in-stent restenosis (Bohme 2021; Brodmann 2013; Del Giudice 2014; Dippel 2015; Gandini 2013; NCT02832024). Eight studies were excluded as they were non-randomised, single-arm trials (Donas 2023; Kim 2023; Lida 2022; NCT02730234; Rundback 2019; Schwindt 2017; Shamma 2020; Shamma 2021). One study was excluded as both arms included atherectomy (Babaev 2022).

See the 'Characteristics of excluded studies' table for reasons for exclusion (Supplementary material 3).

Ongoing studies

We identified seven ongoing studies comparing atherectomy with balloon angioplasty (NCT01579123 [66]; NCT02517827 [67]; NCT02840786 [68]; NCT03380650 [69]; NCT05307263 [70]; NCT05670171 [71]; NCT05868564 [72]). Three studies previously identified as ongoing are now included studies (Rastan 2021; Shamma 2022; Zeller 2022). One study previously identified as ongoing is now listed as excluded, as it compares two different types of atherectomy device (Babaev 2022).

See the 'Characteristics of ongoing studies' table (Supplementary material 4).

Risk of bias in included studies

The risk of bias assessments are presented in the 'Characteristics of included studies' table (Supplementary material 2) and summarised in Figure 2 and Figure 3.

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

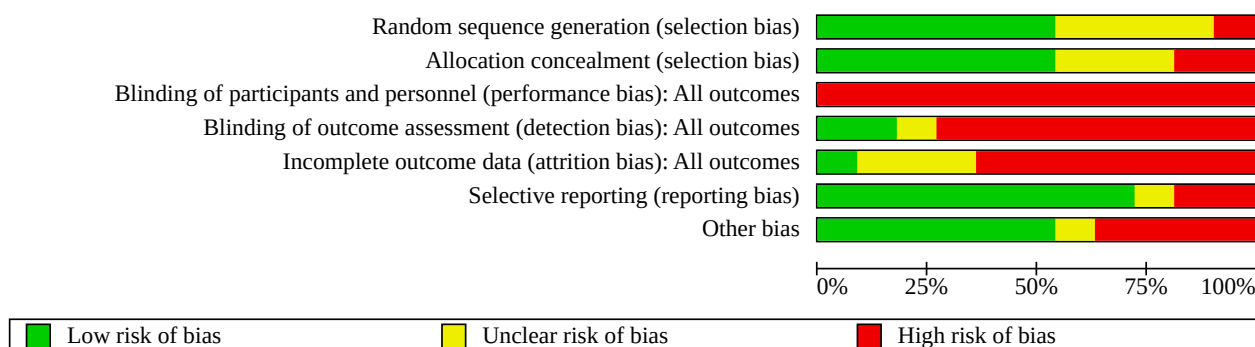


Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Cai 2020	−	−	−	−	?	+	+
Dattilo 2014	?	+	−	−	−	+	?
Nakamura 1995	+	−	−	−	+	?	−
Ott 2017	+	+	−	+	−	+	+
Rastan 2021	+	?	−	+	?	+	+
Shammas 2011	?	+	−	−	−	−	+
Shammas 2012	?	+	−	−	−	+	−
Shammas 2022	+	?	−	−	−	−	−
Vroegindeweij 1995	+	+	−	−	−	+	+
Zeller 2017	+	?	−	?	?	+	+
Zeller 2022	?	+	−	−	−	+	−

We judged six of the studies to be at low risk of bias for randomisation methods: Nakamura 1995 used a random numbers table; Ott 2017 and Rastan 2021 used a computer-generated sequence; Vroegindeweij 1995 used numbered envelopes opened sequentially; Zeller 2017 used block randomisation by centre. Shammas 2022 used a random number generator. Four studies were at unclear risk of bias for randomisation methods: Dattilo 2014 and Zeller 2022 did not state the method of randomisation; Shammas 2011 performed simple randomisation on a 1:1 basis, but did not describe the method of sequence generation; Shammas 2012 stated that sealed envelopes were provided to all centres for randomisation, but did not report the randomisation method. We deemed Cai 2020 at high risk of bias as neither randomisation nor allocation methods were defined.

We deemed six studies to be at low risk for allocation concealment bias. Dattilo 2014, Shammas 2012, Vroegindeweij 1995 and Ott 2017 performed randomisation only after passing the guidewire and used sealed envelopes to conceal allocation. Shammas 2011 and Zeller 2022 used sealed envelopes for allocation concealment. Zeller 2017, Rastan 2021 and Shammas 2022 assigned participants to treatment groups after successful passage of the guidewire across the target lesion, so we judged these studies to be at unclear risk. Cai 2020 and Nakamura 1995 did not report any method of allocation concealment, so we judged these to be at high risk.

It is not possible to blind operators for this procedure, so we assessed all trials to be at high risk for performance bias (Cai 2020; Dattilo 2014; Nakamura 1995; Ott 2017; Rastan 2021; Shammas 2011; Shammas 2012; Shammas 2022; Vroegindeweij 1995; Zeller 2017; Zeller 2022). We also deemed bailout stenting to be at high risk of bias, given that the decision is made at the time of intervention by non-blinded technicians. We deemed two studies at low risk of detection bias. Ott 2017 blinded the independent core laboratory to treatment assignment. Rastan 2021 blinded the study staff and the core laboratory to treatment allocation. Zeller 2017 blinded the duplex ultrasound core laboratory staff, but none of the other outcome assessors, giving an unclear risk of detection bias. We judged all of the remaining studies to be at high risk of detection bias (Cai 2020; Dattilo 2014; Nakamura 1995; Shammas 2011; Shammas 2012; Shammas 2022; Vroegindeweij 1995; Zeller 2022).

Seven studies had a high risk of attrition bias, with six of these being due to significant numbers of participants not being followed up to both six and 12 months (Dattilo 2014; Ott 2017; Shammas 2011; Shammas 2012; Vroegindeweij 1995; Zeller 2022). Shammas 2022 had 255 participants planned for enrolment into the study, but this was significantly affected by the possible association between paclitaxel-coated devices and mortality, and the COVID-19 pandemic [34]. As a result, only 47 participants were included, and the results are considered exploratory. We judged Cai 2020 to be at unclear risk of bias as there was no reporting of loss to follow-up. We judged Zeller 2017 to be at unclear risk of bias. Although participants in the study were lost to follow-up, only 15/102 failed to provide primary outcome data. We judged Rastan 2021 as at unclear risk, as 73/80 participants provided data on the primary outcome. We deemed Nakamura 1995 to be at low risk of bias because there was a complete data set up to six months. Overall, we had serious concerns about the presence of attrition bias.

All studies reported the primary outcomes fully, but three studies failed to completely report all secondary outcomes. Nakamura

1995 reported initial and six-month patencies, but only reported ABIs for participants whose vessels remained patent. We therefore judged this study to be at unclear risk of selective reporting bias. Shammas 2011 failed to completely report follow-up ABIs and did not fully report major adverse events, so we considered this to be at high risk of selective reporting bias. Shammas 2022 failed to report on acute stent thrombosis as part of the major adverse event rate or changes in the walking impairment questionnaire and Rutherford criteria at follow-up. We judged this study to be at high risk of bias. The remaining studies reported all outcomes fully, so we rated them at low risk of selective reporting bias (Cai 2020; Dattilo 2014; Ott 2017; Rastan 2021; Shammas 2012; Vroegindeweij 1995; Zeller 2017; Zeller 2022).

Antiplatelet protocols were clear and uniform in six studies, reducing the risk of confounding by medication differences and so were at low risk of other bias (Cai 2020; Ott 2017; Rastan 2021; Shammas 2011; Vroegindeweij 1995; Zeller 2017). Nakamura 1995, Shammas 2012, Shammas 2022 and Zeller 2022 did not address antiplatelet protocols, which may have impacted outcomes between participant groups. We judged these studies to be at high risk of bias as no antiplatelet protocol was in place. We considered Dattilo 2014 to be at unclear risk of other bias, as the trial randomised vessels rather than participants, meaning a participant could be enrolled more than once and therefore could confound the results.

Synthesis of results

See [Summary of findings 1](#) for the comparison 'Atherectomy compared to balloon angioplasty for peripheral arterial disease'.

See [Summary of findings 2](#) for the comparison 'Atherectomy compared to balloon angioplasty with primary stenting for peripheral arterial disease'.

See [Supplementary material 8](#) for the comparison 'Atherectomy compared to drug-eluting balloon angioplasty for peripheral arterial disease'.

See [Supplementary material 9](#) for the comparison 'Atherectomy compared to drug-eluting balloon angioplasty and primary stenting for peripheral arterial disease'.

We performed meta-analyses using a random-effects model as there was clinical heterogeneity between the studies due to the different devices used ([Supplementary material 5](#); [Supplementary material 6](#)).

Atherectomy versus BA

Primary outcomes

Primary vessel patency

Six studies comparing atherectomy versus BA reported primary patency at six months (Nakamura 1995; Rastan 2021; Shammas 2022; Vroegindeweij 1995; Zeller 2017; Zeller 2022). Atherectomy use may have little to no effect on primary patency compared to BA at six months (RR 1.24, CI 0.92 to 1.68; 6 studies, 298 participants; very low-certainty evidence; Analysis 1.1) or 12 months (RR 1.13, 95% CI 0.96 to 1.34; 5 studies, 326 participants; very low-certainty evidence; Analysis 1.2), but the evidence is very uncertain.

We downgraded the certainty of the evidence from high to very low due to the risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial sizes).

All-cause mortality

Seven studies on atherectomy versus BA reported on mortality rates at one year (Cai 2020; Rastan 2021; Shammas 2011; Shammas 2012; Shammas 2022; Zeller 2017; Zeller 2022). In Shammas 2012, there was an unexpectedly high number of deaths in the BA arm (6/25 (24%) participants), with no deaths in the atherectomy arm, though the trialists could find no good explanation for this. Shammas 2011 reported 4/29 (14%) deaths in the BA arm and 2/29 (7%) deaths in the atherectomy arm. Zeller 2017 reported one-year mortality as 2/48 (4%) deaths in the atherectomy arm compared to 1/54 (2%) deaths in the BA arm. These deaths were not attributed to the procedure, however, with the causes listed as heart failure/stroke, acute coronary syndrome, respiratory failure and neoplastic disorder. Cai 2020 reported 2/49 (4.1%) deaths in the BA arm. The causes of death were cardiac failure for one participant and unknown for the other. There were no reported deaths in the atherectomy group. Rastan 2021 reported a mortality of 3/40 (7.5%) in the atherectomy group and 8/40 (20%) in the BA group. Causes of death included major cardiac events in 3/11, sepsis in 2/11, cancer in 1/11 and unclear in 5/11. The distribution of these causes across the two groups was not reported. Zeller 2022 had a mortality rate of 3/32 (9.4%) in the atherectomy group and 4/34 (11.8%) in the BA group. No causes of death were given. In Shammas 2022 no deaths occurred at one year. Meta-analysis identified no evidence of a difference between the two groups (RR 0.50, 95% CI 0.24 to 1.02; 7 studies, 493 participants; very low-certainty evidence; Analysis 1.3).

We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial sizes).

Fatal and non-fatal cardiovascular events

Five studies reported on cardiovascular outcomes (Cai 2020; Ott 2017; Rastan 2021; Shammas 2011; Zeller 2017). In the atherectomy versus BA comparison, Zeller 2017 reported cardiac failure and acute coronary syndrome as causes of death at 24 months, but it was unclear which participants in which arms these related to. Shammas 2011 declared embolic stroke and myocardial infarction to be secondary outcomes, but recorded no events in either arm. Cai 2020 reported one death (1/49 (2%)), as a result of cardiac failure in the BA group. Rastan 2021 reported that 2/34 (6%) in the atherectomy group had a myocardial infarction, whilst 3/35 (9%) in the BA group had either a myocardial infarction or stroke. Pooled analysis from these two studies identified that there was little to no difference in cardiovascular outcomes between atherectomy and BA (RR 0.59, 95% CI 0.13 to 2.70; 2 studies, 163 participants; very low-certainty evidence; Analysis 1.4). Both trials used DEB as their control groups.

We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial sizes).

Secondary outcomes

Immediate procedural and angiographic outcomes

All studies apart from Rastan 2021 reported on initial technical failure rates. The total pooled analysis identified no evidence of a

difference in the atherectomy group compared to the BA group (RR 0.58, 95% CI 0.29 to 1.17; 9 studies, 637 treated vessels; very low-certainty evidence; Analysis 1.5).

We downgraded the certainty of the evidence from high to very low across both comparisons due to risk of bias (inadequate blinding and high rates of attrition), imprecision (small trial sizes) and inconsistency (from heterogeneity).

Six studies that compared atherectomy with BA reported rates of bailout stenting. Indications identical to all studies were the presence of severe dissection or > 30% residual stenosis (Dattilo 2014; Rastan 2021; Shammas 2011; Shammas 2012; Shammas 2022; Zeller 2017). Shammas 2012 also reported perforation or significant vessel recoil as a reason for bailout stenting. Similarly, Rastan 2021 reported perforation as a reason for bailout stenting. Atherectomy may result in a slight reduction in bailout stenting, but the evidence is very uncertain (RR 0.29, 95% CI 0.11 to 0.80; 6 studies, 442 treated vessels; very low-certainty evidence; Analysis 1.6).

We downgraded the certainty of the evidence from high to very low due to the risk of bias (inadequate blinding and high rates of attrition), imprecision (small trial sizes) and inconsistency (from heterogeneity).

Four studies reported balloon inflation pressures (Dattilo 2014; Shammas 2011; Shammas 2012; Shammas 2022). Our pooled meta-analysis identified that atherectomy may result in a slight reduction in balloon inflation pressures compared to BA, but the evidence is very uncertain (MD -2.77, 95% CI -4.80 to -0.74; 4 studies, 260 treated vessels; very low-certainty evidence; Analysis 1.7). A difference between subgroups was identified ($\text{Chi}^2 = 12.39$, $\text{df} = 1$ ($P = 0.0004$), $I^2 = 91.9\%$). The use of atherectomy may result in lower balloon inflation pressures required to achieve optimal inflation in subsequent angioplasty, compared to PBA alone (MD -3.68, 95% CI -5.36 to -2.01; 3 studies, 213 treated vessels; very low-certainty evidence; Analysis 1.7), whereas we did not identify a difference in balloon inflation pressures in the atherectomy plus DEB versus DEB subgroup (MD -0.10, 95% CI -1.18 to 0.98; 1 study, 47 treated vessels; very low-certainty evidence; Analysis 1.7). The high heterogeneity between studies and the minimal data available for balloon inflation pressures in the atherectomy plus DEB versus DEB subgroup reduces our confidence in this finding.

We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition), imprecision (small trial sizes) and inconsistency (from heterogeneity).

Target vessel revascularisation rates

There was some variation between trials in whether they collected target lesion revascularisation data or TVR data, or both. The previous authors made the decision to collect TVR data, to allow the fairest comparison between all studies, and we have continued with the same methodology. Dattilo 2014 reported target lesion revascularisation, hence this was excluded from the analysis. Zeller 2022 reported target lesion revascularisation, with a higher total number of lesions than participants. This study was also excluded.

In this update, we identified three new studies that we have included in our analysis (Cai 2020; Rastan 2021; Shammas 2022). All three studies reported target lesion revascularisation rates,

however the number of lesions was the same as the number of participants and, as a result, we included data from these studies in the analysis.

In total, five studies comparing atherectomy versus BA reported TVR at six months (Cai 2020; Rastan 2021; Shammas 2012; Shammas 2022; Zeller 2017). The analysis found no evidence of a difference in TVR rates in the atherectomy versus BA comparison (RR 0.61, 95% CI 0.24 to 1.56; 5 studies, 348 participants; very low-certainty evidence; Analysis 1.8). We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial sizes).

At 12 months, six studies reported TVR (Cai 2020; Rastan 2021; Shammas 2011; Shammas 2012; Shammas 2022; Zeller 2017). The pooled analysis identified no evidence of a difference between the atherectomy versus BA arms (RR 0.68, 95% CI 0.41 to 1.12; 6 studies, 371 participants; very low-certainty evidence; Analysis 1.9). We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial sizes).

Complication rates

Seven studies in the atherectomy versus BA comparison reported an overall complication rate (Dattilo 2014; Nakamura 1995; Shammas 2011; Shammas 2012; Vroegindeweij 1995; Zeller 2017; Zeller 2022), with no evidence of a difference detected in our meta-analysis between the two arms (RR 0.84, 95% CI 0.34 to 2.04; 7 studies, 457 participants; very low-certainty evidence; Analysis 1.10).

We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition), imprecision (small trial sizes) and inconsistency (from heterogeneity).

Nine studies reported on embolisation rates (Cai 2020; Dattilo 2014; Nakamura 1995; Shammas 2011; Shammas 2012; Shammas 2022; Vroegindeweij 1995; Zeller 2017; Zeller 2022). We identified no evidence of a difference between atherectomy versus BA groups (RR 1.86, 95% CI 0.53 to 6.51; 9 studies, 598 participants; very low-certainty evidence; Analysis 1.11). Seven studies in the atherectomy versus BA comparison reported on dissection rates (Cai 2020; Dattilo 2014; Shammas 2012; Shammas 2022; Vroegindeweij 1995; Zeller 2017; Zeller 2022). Pooled analysis identified that atherectomy use may result in a slight reduction in dissection rates compared to BA, however the evidence is very uncertain (RR 0.38, 95% CI 0.17 to 0.84; 7 studies, 501 participants; very low-certainty evidence; Analysis 1.12). We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition), imprecision (small trial sizes) and inconsistency (from heterogeneity).

Details of specific complications in each individual included study are provided in [Supplementary material 7](#).

Morbidity assessment

Seven studies reported rates of amputation (Cai 2020; Ott 2017; Rastan 2021; Shammas 2011; Shammas 2012; Shammas 2022; Zeller 2017). Notably, Rastan 2021 reported an amputation rate significantly higher than any other trial. The trialists reported on

unplanned major or minor amputation of the target limb, with an amputation rate of 8/37 (21.6%) in the atherectomy group and 11/35 (31.4%) in the BA group. Shammas 2011 was the only other trial to report an amputation. The trialists reported one amputation in the BA group (1/24 (4.2%)). No other trial reported an amputation event. Shammas 2011 and Cai 2020 do not specify whether the amputation rate relates to major or minor amputations. Shammas 2012 reported on device or procedure-related above ankle amputations. Shammas 2022 reported amputation rate as part of major adverse events at one year and included major amputations of the treated limb, however in the two-year exploratory results available, the trialists reported no major or minor amputation in either group.

Our analysis identified no evidence of a difference in amputation rates when comparing atherectomy versus BA (RR 0.66, 95% CI 0.31 to 1.41; 6 studies, 391 participants; very low-certainty evidence; Analysis 1.13). We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial sizes).

Quality of life outcomes

None of the included studies reported on quality of life outcomes.

Clinical and symptomatic outcomes

Four atherectomy versus BA trials reported clinical and symptomatic outcomes. Zeller 2017 reported functional outcomes in their study, including pain score, walking distance score, walking speed score and stair climbing score for baseline, six months and one year. However, they did not report any benefit for any of the outcomes at either time period. Shammas 2011 reported 30-day and 12-month ABI and Rutherford class, and stated that there was no difference between any of these outcomes in the two treatment arms. Rastan 2021 reported on Rutherford classification, ankle brachial index, toe brachial index and walking impairment questionnaire scores. No significant difference was found between the two groups at each interval follow-up. Zeller 2022 reported on changes in Rutherford criteria, with no significant difference between groups at each interval follow-up.

Femoropopliteal subgroup

Six atherectomy versus BA trials focused on lesions within the femoropopliteal segment (Cai 2020; Dattilo 2014; Nakamura 1995; Shammas 2022; Vroegindeweij 1995; Zeller 2017). For lesions in the femoropopliteal segment, the use of atherectomy may lead to a reduction in dissection rates (RR 0.38, CI 0.17 to 0.83; 5 studies, 381 participants; very low-certainty evidence; Analysis 3.12) and bailout stenting rates (RR 0.07, 95% CI 0.02 to 0.22; 3 studies, 214 participants; very low-certainty evidence; Analysis 3.6) when compared to angioplasty alone, however the evidence is very uncertain. There was no evidence of a difference between the two arms in any of the other outcome measures.

Atherectomy versus stenting

Primary outcomes

Primary vessel patency

In the atherectomy versus stenting study, Ott 2017 did not report primary patency.

All-cause mortality

In the atherectomy versus stenting comparison, Ott 2017 reported mortality at six and 24 months as a secondary outcome. Ott 2017 reported one postprocedural death caused by haemorrhagic shock secondary to retroperitoneal bleeding in the stenting and drug-eluting balloon arm. At 24 months, they reported three deaths in the DEB plus stent arm, one death in the PBA plus stent arm and no deaths in the atherectomy arm. All of these deaths were attributed to underlying cardiovascular disease. In the atherectomy versus stenting analysis, we identified no evidence of a difference in mortality rates (RR 0.38, 95% CI 0.04 to 3.23; 1 study, 155 participants; very low-certainty evidence; Analysis 2.1). We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial size).

Fatal and non-fatal cardiovascular events

In the atherectomy versus stenting comparison, Ott 2017 reported four deaths at 24 months (three deaths in the DEB plus stent arm and one death in the PBA plus stent arm). The trialists attributed these to underlying cardiovascular disease, but did not specify any causes. In the atherectomy versus stenting analysis, there was no evidence of a difference in fatal and non-fatal cardiovascular events (RR 0.38, 95% CI 0.04 to 3.23; 1 study, 155 participants; very low-certainty evidence; Analysis 2.2). We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial size).

Secondary outcomes

Immediate procedural and angiographic outcomes

In the atherectomy versus stenting trial (Ott 2017), there were no initial technical failures in either arm (Analysis 2.3).

The atherectomy versus stenting trial by Ott 2017 reported that 14/55 (25%) participants in the atherectomy group received bailout stenting due to flow-limiting dissections, one of whom developed thrombus requiring thrombus aspiration. Two participants had perforations, one treated with a covered stent, the other by prolonged balloon inflation and protamine administration.

Target vessel revascularisation rates

For the comparison of atherectomy versus stenting, Ott 2017 reported TVR at six months and 24 months. There was no evidence of a difference between the treatment arms for either six-month TVR (RR 2.27, 95% CI 0.95 to 5.46; 1 study, 155 participants; very low-certainty evidence; Analysis 2.4) or 24-month TVR (RR 2.05, 95% CI 0.96 to 4.37; 1 study, 155 participants; very low-certainty evidence; Analysis 2.5). We downgraded the certainty of the evidence from high to very low due to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial size and wide confidence intervals).

Complication rates

In the atherectomy versus stenting comparison, Ott 2017 reported three complications, all of which were in the atherectomy arm. The meta-analysis did not identify a difference between treatment arms for complication rate (RR 7.04, 95% CI 0.80 to 62.23; 1 study, 155 participants; very low-certainty evidence; Analysis 2.6). We downgraded the certainty of the evidence from high to very low due

to risk of bias (inadequate blinding and high rates of attrition) and imprecision (small trial size and wide confidence intervals).

Morbidity assessment

There were no amputation events in the study that compared atherectomy versus stenting (Analysis 2.7; Ott 2017).

Quality of life outcomes

The atherectomy versus stenting trial did not report any quality of life outcomes (Ott 2017).

Clinical and symptomatic outcomes

The atherectomy versus stenting trial did not report clinical and symptomatic outcomes (Ott 2017).

Atherectomy versus bypass procedure

We found no studies comparing atherectomy to an open bypass procedure.

Sensitivity analysis

Four studies treated more than one vessel per participant or limb (Dattilo 2014; Shammass 2011; Shammass 2012; Zeller 2022). It is possible, therefore, that the outcomes of these trials received greater weight in the meta-analysis than is appropriate. Ideally, we would have carried out a sensitivity analysis to assess the impact this had on the results. However, given the low number of studies, we felt that this would not add meaningful results.

Only Vroegindeweij 1995 did not perform routine angioplasty with atherectomy. We performed sensitivity analysis to assess the effect of including this study in the overall meta-analyses of the primary outcomes (primary vessel patency and all-cause mortality) and did not observe any difference between including or excluding this study in the analysis.

DISCUSSION

Summary of main results

This updated review includes 11 RCTs (814 participants and 872 treated lesions) and identified two comparisons, atherectomy versus BA and atherectomy versus stenting.

In the atherectomy versus BA comparison (10 studies, 659 participants), there was no evidence of a difference in primary patency at six and 12 months, mortality rates, fatal and non-fatal cardiovascular events, target vessel revascularisation at six and 12 months, and overall complication rates. See [Summary of findings 1](#).

In the atherectomy versus stenting comparison (1 study, 155 participants), primary patency was not reported. There was no evidence of a difference in mortality rates, fatal and non-fatal cardiovascular events, target vessel revascularisation at six months, and overall complication rates. See [Summary of findings 2](#). Target vessel revascularisation at 12 months was not reported.

For both comparisons, across all outcomes, the evidence was very uncertain.

Limitations of the evidence included in the review

All 11 included studies were of poor methodological quality, with a high risk of overall bias due to a lack of blinding and high attrition,

meaning the conclusions that can be drawn from the analyses are severely limited. There was significant statistical heterogeneity between studies due to the small participant numbers. In addition, there was heterogeneity due to clinical differences in participant groups, trial protocols and target vessels. Only one included trial compared stenting versus atherectomy (Ott 2017). We interpreted the results for all outcomes with caution, given the small trial sizes and lack of information. Using the GRADE approach, we judged all outcomes to have very low-certainty evidence [40]. [Summary of findings 1](#) and [Summary of findings 2](#) show that there is currently no clear evidence to support the use of atherectomy as a treatment for peripheral vascular disease. [Supplementary material 8](#) and [Supplementary material 9](#) show that the use of drug-eluting technology has little to no effect on outcomes. We downgraded all the outcomes from high to very low certainty due to risk of bias (inadequate blinding and high rates of attrition), imprecision (small trial sizes) and inconsistency (from heterogeneity).

Only Ott 2017, Rastan 2021 and Shammass 2011 utilised power calculations to assess the required number of participants. Overall study numbers were very low, and meta-analysis of very low participant numbers in randomised trials can be unreliable [73]. As a result, the observed lack of difference in primary patency could easily be a type II error. Four of the trials did not state medication protocols (Nakamura 1995; Shammass 2012; Shammass 2022; Zeller 2022). This may be important as it is known that the use of antiplatelet agents, cilostazol and heparin are all associated with lower restenosis rates after angioplasty [74]. The included trials did not include important clinical endpoints such as secondary patency, limb survival and complication rates between techniques in sufficient detail.

In contemporary clinical practice, atherectomy is not used as a standalone treatment, but instead prior to adjunctive balloon angioplasty with or without stenting. Only one included study, which is 30 years old, compared atherectomy alone without adjunctive treatment to BA (Vroegindeweij 1995). Vroegindeweij 1995 was included in this review as it met the predefined eligibility criteria; however, its results have limited applicability to modern clinical practice and hence we performed a sensitivity analysis to assess its inclusion in the meta-analysis. Three participants in the atherectomy arm of Vroegindeweij 1995 crossed over and had subsequent BA after failure of atherectomy alone.

Open bypass surgery remains a treatment option in current practice, and the lack of identification of a direct comparison is a further limitation to the evidence around atherectomy use.

This update includes all information from RCTs identified by the searches, and we have presented an up-to-date meta-analysis of atherectomy versus any other therapy for peripheral arterial disease (PAD).

The indication for intervention was claudication in two studies (Nakamura 1995; Vroegindeweij 1995), claudication or rest pain in two studies (Dattilo 2014; Zeller 2017) and claudication, rest pain or tissue loss in seven studies (Cai 2020; Ott 2017; Rastan 2021; Shammass 2011; Shammass 2012; Shammass 2022; Zeller 2022). Results of angioplasty and bypass surgery are known to vary between people with these indications, so this may bias the results from the different studies [33]. Some of the studies did not state the severity of claudication, which may mean that the participants would have been managed conservatively in many centres, so

the results should be interpreted with a degree of caution [75]. Unfortunately, we were not able to separate results by symptoms (claudication or chronic limb-threatening ischaemia) because of the way the studies reported results. In addition, the majority of included studies did not report on all of this review's prespecified outcomes. Therefore, the findings of this review are based in most cases on results from a minority of the included studies.

Amputation-free survival is an important endpoint in trials of PAD, but the included studies did not report this. One reason may be the reliance on including participants with a less severe form of peripheral arterial disease (claudication) in the trials.

Mortality is commonly reported in trials of lower limb revascularisation, which is why we considered it a primary outcome measure. However, mortality rates from angioplasty are much lower than primary patency or limb loss rates, so trials would not be expected to show a difference if powered to detect primary patency differences [76, 77]. The results presented may be a consequence of random error due to few events in small sample sizes. There was little difference in all-cause mortality rates between interventions, but mortality was only reported in eight of the 11 included trials, and rates of death were low.

There were also differences in the modality of follow-up between studies: Ott 2017 used angiogram. Rastan 2021 used angiography and duplex ultrasound. Similarly, Cai 2020, Dattilo 2014, Shammass 2022, Vroegindeweij 1995 and Zeller 2022 and used duplex ultrasound. Zeller 2017 used duplex ultrasound and plain ultrasound. Nakamura 1995 used doppler pressures. Shammass 2011 and Shammass 2012 did not use imaging at follow-up. These differences in outcome collection should be considered when interpreting the results.

Limitations of the review processes

Despite carrying out a thorough unrestricted search, our review process identified only 11 trials of varying size, so it is difficult to assess the impact of reporting bias.

Four trials treated more than one vessel per participant or limb (Dattilo 2014; Shammass 2011; Shammass 2012; Zeller 2022). Failure of patency of any of the treated vessels increases the chances that other treated vessels will cease to be patent, so these observations will be correlated. It is possible, therefore, that the outcomes of these trials are given greater weight in the meta-analysis than is appropriate in the analysis of six-month and 12-month patency. As both the angioplasty and atherectomy arms of these trials included multiple vessels per participant, it is unlikely that the magnitude of the observed effect has been affected significantly, though our degree of confidence in this effect may be overstated.

In regard to the GRADE domain of indirectness, we identified some discrepancies between planned outcome measures and what was reported in studies. There was variation in whether trials collected target lesion revascularisation data or TVR data, or both. Upon discussion, we made the decision to collect TVR, as we felt this allowed us to perform the fairest comparison between all studies. Similarly, there was variation between trials in outcome definitions, with studies collecting primary patency rates or occlusion rates. Given the inter-trial variation for the definitions between the two, we only included studies that collected 'primary patency' in the meta-analysis. Furthermore, we were unable to identify any

head-to-head comparison comparing atherectomy to open bypass surgery.

The results are applicable to the overall population with PAD; however, one major limitation is the lack of TASC II classifications in studies, meaning the results may be representative of specific subgroups of lesions.

Agreements and disagreements with other studies or reviews

This is an update of a previous Cochrane review of atherectomy for PAD [9]. The previous review included six RCTs investigating atherectomy versus BA and one RCT comparing atherectomy to primary stenting. The review authors reported that atherectomy use may reduce dissection and bailout stenting rates. No evidence of differences was identified between the DEB and PBA subgroups. In this update, we similarly report that atherectomy use may reduce the rate of dissections and bailout stenting compared to BA; however, the evidence for this is very uncertain. Our analysis also identified that the use of atherectomy may reduce the balloon inflation pressure required for subsequent BA, compared to BA alone; however, again, this is based on very uncertain evidence. Subgroup analysis identified lower balloon inflation pressures when comparing atherectomy to PBA, but not for DEB. However, this finding is very uncertain. The previous review authors stated that atherectomy use may increase embolisation rates; however, in our update, with the addition of four new studies, we did not identify a difference in embolisation rates comparing atherectomy to BA. This may be as a result of improvements in atherectomy devices or techniques in more recent studies, causing a reduction in rates of this complication.

Since the publication of the previous Cochrane review on atherectomy use in 2020, there have been a number of relevant systematic reviews published in the literature [15, 16, 78]. All three reviews compared atherectomy to balloon angioplasty, with two focusing on specific anatomical locations, either femoropopliteal or infrapopliteal disease [15, 16]. Two reviews included data from observational studies, which made up the majority of data in the meta-analyses (7/10 studies [16], 10/15 studies [78]). Both reviews including observational data identified a reduction in target lesion revascularisation and amputation rates in the atherectomy group. One review included RCTs comparing atherectomy to BA for femoropopliteal disease [15]. Despite the inclusion of a number of trials that we have included in this review, the authors identified an increased primary patency and lower amputation rates in the atherectomy group. We completed a post hoc analysis investigating atherectomy compared to BA in the femoropopliteal segment. We identified that atherectomy use may result in lower rates of bailout stenting and dissection; however, these findings are very uncertain, and therefore not in agreement with the above findings from Dikang et al. We used recommended Cochrane techniques to extract data from studies and Kaplan-Meier curves, whereas the meta-analysis by Dikang et al. appears to have used a different technique, as the extracted numbers are different [18].

The use of balloon angioplasty and stenting for peripheral vascular disease is widely practised. Most recently, the multi-centre BASIL-2 trial comparing vein bypass to angioplasty and stenting reported a reduction in mortality and an associated increase in amputation-free survival with an endovascular first approach [79]. Of note,

atherectomy was permitted, but not used in the endovascular arm of this trial.

The use of paclitaxel-coated balloons or stents has faced controversy since a 2018 meta-analysis identified an association between drug-coating technology and mortality risk [34]. A 2022 meta-analysis of 59 studies identified no evidence of a difference in mortality risk with paclitaxel-coated devices compared to uncoated balloons or stents at one, two and five-year intervals [80]. Similarly, in our meta-analysis, there was no evidence of increased mortality risk with the drug-eluting BA subgroup; however, our results are very uncertain.

AUTHORS' CONCLUSIONS

Implications for practice

This updated review shows that the evidence is very uncertain about the effect of atherectomy on primary patency, mortality and cardiovascular event rates compared to balloon angioplasty with or without stenting. However, there appears to be little to no effect on these outcomes based on the data. We did not detect any evidence of differences in target vessel revascularisation rates or complication rates; however, the evidence is very uncertain. The included studies were small, heterogeneous and at high risk of bias. Larger studies powered to detect clinically meaningful, patient-centred outcomes are still required.

Implications for research

Current evidence in this area is still limited. Larger and better designed trials in selected subgroups of participants are needed to increase our confidence in the evidence.

Future trials should address the following factors.

- Adequately powered randomised controlled trials should be designed with patient-centred outcomes. Patency rates should be avoided as primary outcomes.
- Novel designs such as platform trials would be useful in this area as there are so many changing treatment options.
- Studies examining surgery versus atherectomy are needed, as we found no randomised controlled trials for this comparison. Participants should be stratified according to whether they suffer from intermittent claudication or chronic limb-threatening ischaemia.
- More rigorous follow-up is needed. It is important that future studies have both longer follow-up and blinded outcome assessment. The procedures are sometimes performed by interventional radiologists, but followed up by vascular surgeons, so this aim is achievable.
- Study designs should include outcomes that are more participant-centred; for example, quality of life, pain scores and psychosocial measures.

We could not include cost in this Cochrane review, but this is a significant factor when considering results from atherectomy, as it is more expensive than balloon angioplasty. Future research should explore this.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD006680.pub4](https://doi.org/10.1002/14651858.CD006680.pub4).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Complications

Supplementary material 8 Summary of findings table 3: Atherectomy compared to drug eluting balloon angioplasty for peripheral arterial disease

Supplementary material 9 Summary of findings table 4: Atherectomy compared to drug-eluting balloon angioplasty and primary stenting for peripheral arterial disease

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Professor Stewart Walsh, Discipline of Surgery, University of Galway, Ireland; Toby Lasserson, Cochrane Editor-in-Chief;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Ben Ridley, Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Jacob Hester, Central Editorial Service;
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Contributions of authors

SP: study selection, assessed trial quality, extracted data, updated the review text.

AG: study selection, assessing trial quality.

SS: literature search.

GA: conceptualisation, reviewed and edited the update.

RH: conceptualisation, reviewed and edited the update.

CT: conceptualisation, supervision, extracted data, reviewed and edited the update.

Some authors involved in the previously published versions of this review in 2014 and 2020 are no longer included in the author byline: Rami Radwan, Bethany Wardle, Paul Hayes, Gautham Viswanathan and Sankar D Navaneethan. Some of the content retained in this review reflects their contributions.

Declarations of interest

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Review update (2020): [10.1002/14651858.CD006680.pub3](https://doi.org/10.1002/14651858.CD006680.pub3)

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the *Cochrane Library*: Full search strategies for each database; full citations of each unique report for all studies included, ongoing or waiting classification, or excluded at the full-text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from (Excel) are available from the authors on reasonable request.

What's new

Date	Event	Description
6 January 2026	New search has been performed	A new search was run, initially on 23 December 2023 with an updated search on 28 January 2025.
6 January 2026	New citation required but conclusions have not changed	New search run. Four new studies were included. Eight new studies were excluded, and seven ongoing studies were identified. Review text updated in keeping with current Cochrane standards, including the addition of two new summary of findings tables. No change to conclusions.

History

Protocol first published: Issue 3, 2007

Review first published: Issue 3, 2014

Date	Event	Description
1 October 2019	New search has been performed	New search run. Three new studies included. Six new studies were excluded and 11 new ongoing studies were identified. Review text updated in keeping with current Cochrane standards including addition of 'Summary of findings' tables. No change to conclusions.
29 October 2008	Amended	Converted to new review format.

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ADDITIONAL TABLES

Table 1. Overview of included studies and synthesis table

Study name, year, country	Study design	Intervention details	Population (sample size: intervention/control)	Primary outcomes	Secondary outcomes
Cai 2020 China	RCT	Atherectomy + DEB vs DEB alone	No of participants: 94 Atherectomy + DEB: 45 DEB alone: 49	Primary patency	Technical success, limb amputation, any death

Table 1. Overview of included studies and synthesis table (Continued)

Datillo 2014 USA	RCT	Atherectomy + PBA vs PBA alone	No. of participants: 50 Atherectomy + PBA: 25 PBA alone: 25	Freedom from TLR	Changes in ABI and Rutherford class
Nakamura 1995 USA	RCT	Atherectomy + PBA vs PBA alone	No. of participants: 39 Atherectomy + PBA: 26 PBA alone: 13	Primary patency	Changes in ABI
Ott 2017 Germany	RCT	Atherectomy + bailout stenting vs DEB + stenting vs PBA + stenting	No. of participants: 155 PBA followed by DEB and stenting: 48 PBA and stenting: 52 Atherectomy: 55	Percentage diameter stenosis after 6 months	TLR Thrombosis Ipsilateral amputation Binary restenosis All-cause mortality
Rastan 2021 Germany and Austria	RCT	Atherectomy + DEB vs DEB alone	No. of participants: 80 Directional atherectomy + DEB: 40 DEB alone: 40	Primary patency	Secondary patency and CD-TLR rate Changes in ABI, TBI, RC and walking impairment questionnaire scores Minor and major amputation rates Death, myocardial infarction, stroke, unplanned major and minor amputation of the target limb
Shammas 2011 USA	RCT	Atherectomy + PBA vs PBA alone	No. of participants: 58 Atherectomy plus PBA: 29 PBA: 29	TLR	Bailout stenting Immediate angiographic results TLR Major adverse events Change in the ABI
Shammas 2012 USA	RCT	Atherectomy + PBA vs PBA alone	No. of participants: 50 Atherectomy plus PBA: 25 PBA: 25	Ability to achieve adequate lumen diameter, defined as < 30% residual stenosis with no bailout stenting or dissection	Bailout stenting Limb salvage TLR and TVR Major adverse events
Shammas 2022 USA	RCT	Atherectomy + DEB vs DEB alone	No. of participants: 47 Atherectomy + DEB: 31 DEB: 16	TLR Major adverse events	Device outcome Procedural outcome Clinical patency Major adverse events

Table 1. Overview of included studies and synthesis table (Continued)

					Change in WIQ score
					Changes in Rutherford clinical category
					Change in ABI
Vroegindeweij 1995 Netherlands	RCT	Atherectomy + PBA vs PBA alone	No. of participants: 73 Atherectomy + PBA: 38 PBA: 35	Primary patency	Restenosis
Zeller 2017 Belgium, Germany, Poland and Switzerland	RCT	Atherectomy + DEB vs DEB alone	No. of participants: 102 Atherectomy + DEB: 48 DEB alone: 54	Target lesion percentage diameter stenosis at 1 year post-procedure	Technical success Major adverse events
Zeller 2022 Germany and Austria	RCT	Atherectomy + DEB vs DEB alone	No. of participants: 66 Atherectomy + DEB: 32 DEB alone: 34	Technical success, device success, primary patency of target lesion	Changes in the Rutherford category, clinically driven target lesion revascularisation, unplanned/unavoidable major amputation, all-cause death and major adverse events

ABI: ankle brachial index; CD-TLR: clinically driven target lesion revascularisation; DEB: drug-eluting balloon angioplasty; No: number; PBA: plain balloon angioplasty; RC: Rutherford criteria; RCT: randomised controlled trial; TLR: target lesion revascularisation; TVR: target vessel revascularisation; WIQ: walking impairment questionnaire

Table 2. Intervention details for included studies

Study	Atherectomy device type (brand)	Control group	Antiplatelet protocol
Cai 2020	Directional atherectomy (Silverhawk or Turbohawk) with distal embolism filter	DEB	Aspirin 100 mg, clopidogrel 75 mg, starting 3 days pre-procedure, continued for 6 months, followed by 1 antiplatelet
Datillo 2014	Orbital (Diamondback) atherectomy	PBA	Single antiplatelet aspirin/clopidogrel pre-procedure and then DAPT for minimum 4 to 6 weeks post-procedure
Nakamura 1995	Directional (transluminal extraction catheter) atherectomy	PBA	Does not specify
Ott 2017	Directional (Silverhawk) atherectomy with distal embolism filter	DEB or PBA plus stenting	500 mg aspirin IV immediately after procedure, then DAPT aspirin/clopidogrel for 6 months, followed by single antiplatelet indefinitely
Shammas 2011	Directional (Silverhawk) atherectomy with distal embolism filter used in 1/2 of participants	PBA	Loading doses of antiplatelet immediately prior to procedure
Shammas 2012	Orbital (Diamondback) atherectomy	PBA	Does not specify
Shammas 2022	Rotational (Jetstream) atherectomy	DEB	Does not specify

Table 2. Intervention details for included studies *(Continued)*

Vroegindeweij 1995	Directional (Simpson) atherectomy	PBA	Low-dose aspirin started 1 day pre-procedure. Length and dose not specified
Zeller 2017	Directional (Silverhawk) with distal embolism filter	DEB	DAPT 4 weeks pre-procedure, continuing to 4 weeks post-procedure, followed by aspirin indefinitely
Zeller 2022	Orbital atherectomy	DEB	Does not specify

DAPT: dual antiplatelets; DEB: drug-eluting balloon angioplasty; IV: intravenous; PBA: plain balloon angioplasty

INDEX TERMS

Medical Subject Headings (MeSH)

Amputation, Surgical [statistics & numerical data]; *Atherectomy [adverse effects] [methods] [mortality]; Humans; Intermittent Claudication [surgery]; Ischemia [surgery]; *Peripheral Arterial Disease [mortality] [surgery] [therapy]; Randomized Controlled Trials as Topic